Unique Signature Schemes with Tight Security in the ROM via Hash-and-Subset-Sign

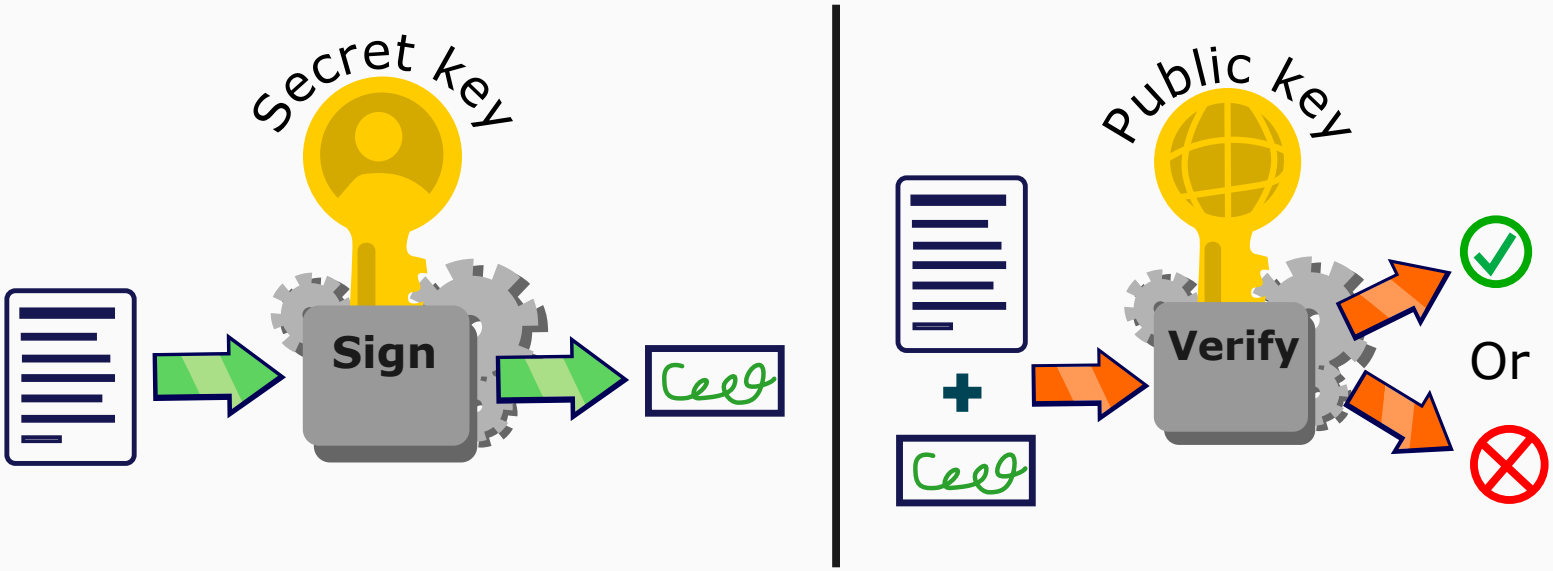
Thomas Attema[1,2], Lisa Kohl[2], **Joost van der Laan**[3]

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[1]TNO, Applied Cryptography & Quantum Applications
[2]Cryptology Group, CWI
[3]TCS Group, University of Amsterdam

Introduction

What are (Unique) Signature Schemes

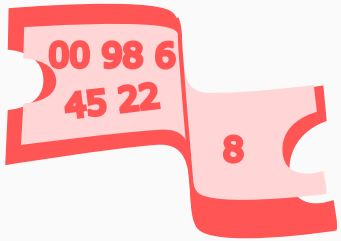


Uniqueness

$\forall \mathsf{pk}, m, \sigma_1, \sigma_2 : \mathsf{Verify}(\mathsf{pk}, \sigma_1, m) = \mathsf{Verify}(\mathsf{pk}, \sigma_2, m) = 1 \implies \sigma_1 = \sigma_2.$

Advantages of Uniqueness

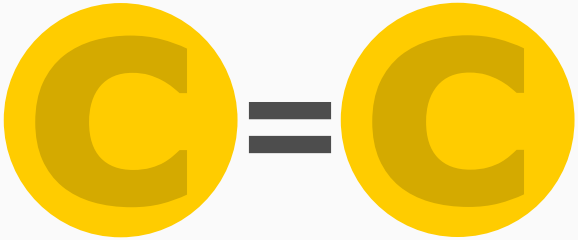
Unique Signatures $\Longrightarrow$
Verifiable Random Functions.



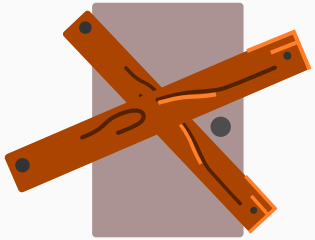
Prevents favourable resampling:
When used to derive an
aggregate seed.



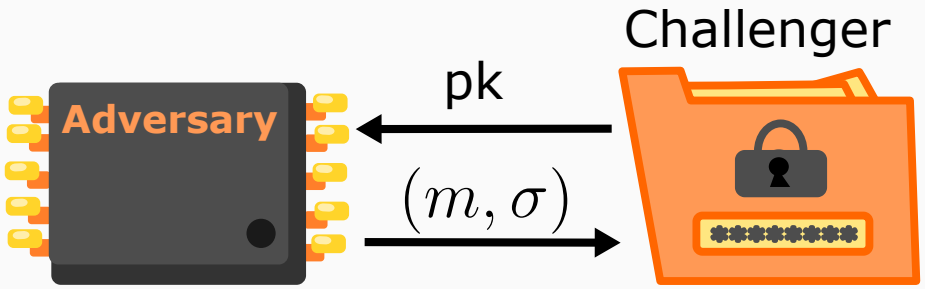
Prevents "double spending"
when used for coins.

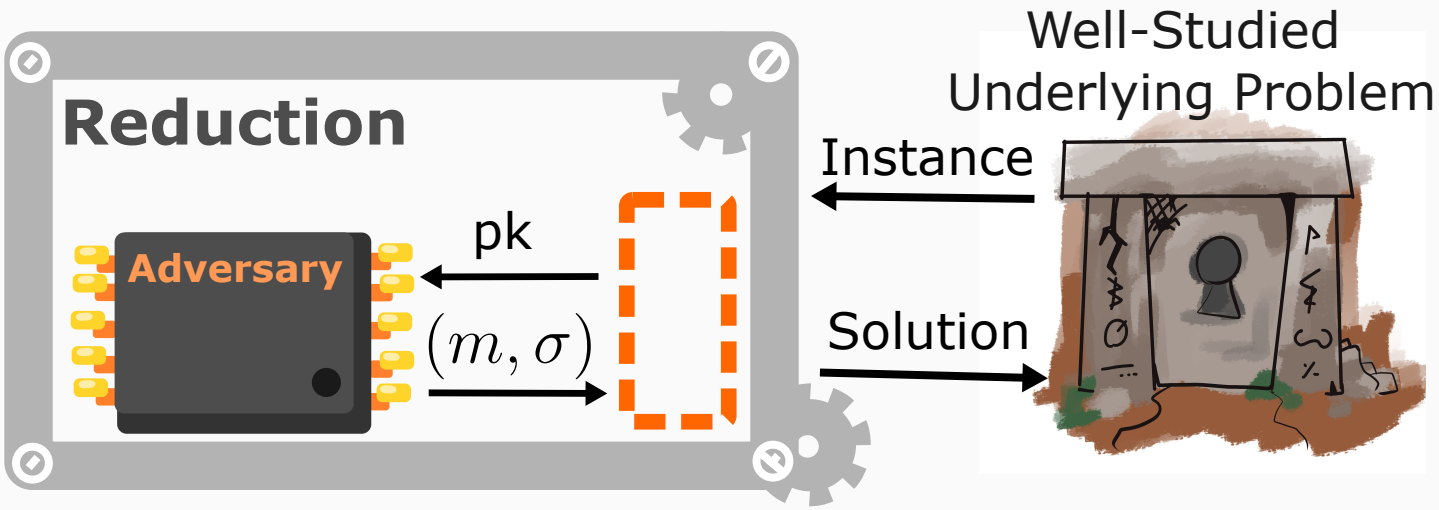


Backdoor resillience: Implementations
cannot reveal secret information
by strategically picking signatures.



Reductions





Security Loss: Ratio of success probabilities $\ell_{\mathcal{R}}(\lambda) = \epsilon_{\mathcal{A}}(\lambda)/\epsilon_{\mathcal{R}^{\mathcal{A}}}(\lambda)$.

Where $\varepsilon_X(\lambda)$: success probability of X.

Tight Security

Reminder

Security Loss: Ratio of success probabilities $\ell_{\mathcal{R}}(\lambda) = \epsilon_{\mathcal{A}}(\lambda)/\epsilon_{\mathcal{R}^{\mathcal{A}}}(\lambda)$.
Where $\varepsilon_X(\lambda)$: success probability of X.

Tight Security

A reduction is tight if $\ell_{\mathcal{R}}(\lambda)$ is independent of λ or the amount of signatures q_{Sign} an attacker has access to.

- Tight security reductions lead to smaller concrete parameters.
- Avoid having to estimate the amount of signatures signed q_{Sign}.

Prior Results

Standard Model: $\Omega(q_{\mathsf{Sign}})$ loss is inherent [Coron02]

Scheme	Unique	Tight	#ROs = const
FDH [BelRog93a]	✓		✓
PSS [KatWan03, BelRog96b]		✓	✓
Chain-based [GCS+17,Sha18]	✓	✓	

Can there exist a tightly secure unique signature scheme which uses a constant number of random oracle queries for signing and verification?

Results

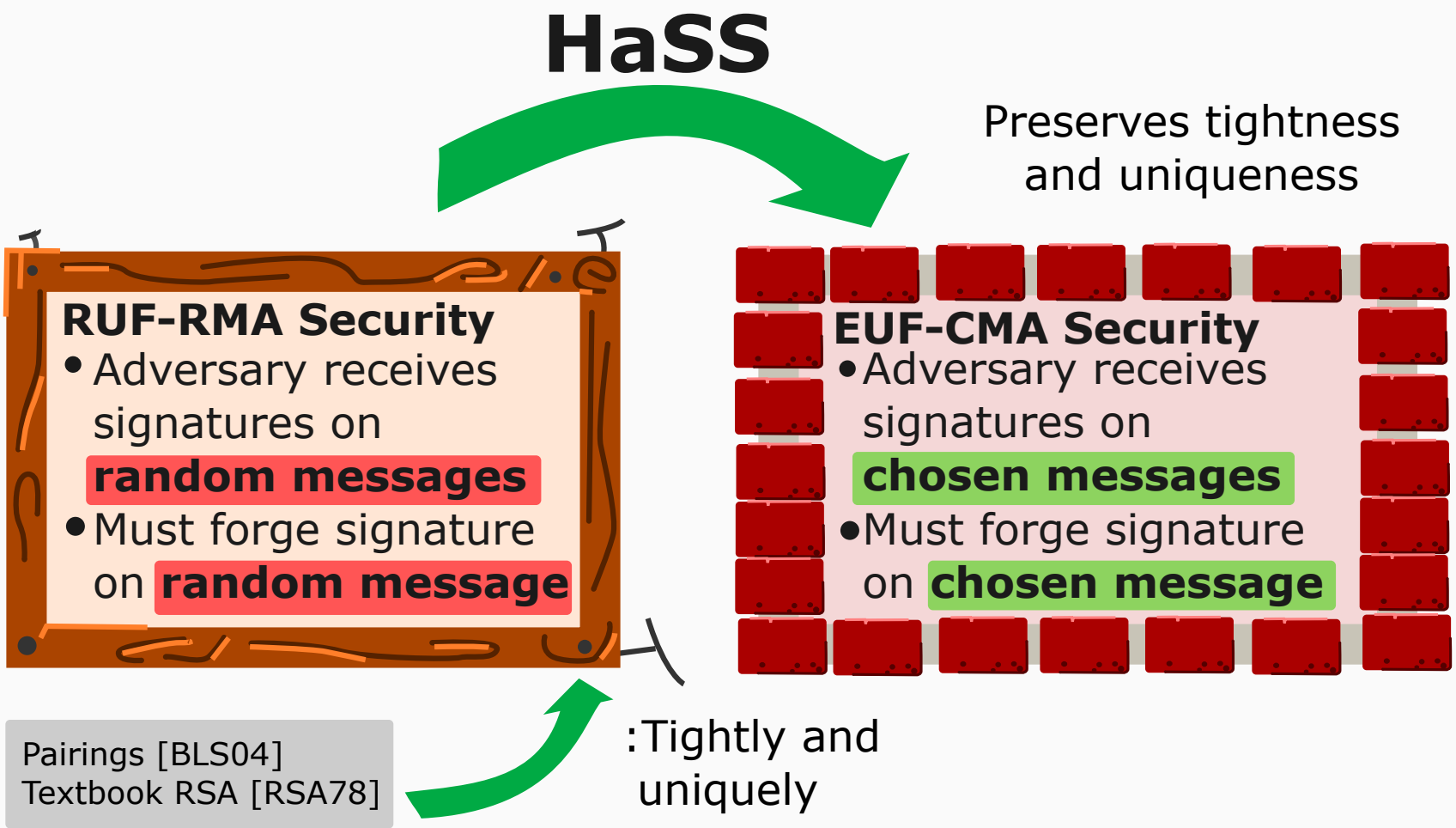
Result

Answer: Yes!

Scheme	Unique	Tight	#ROs = const
FDH [BelRog93]	✓		✓
PSS [KatWan03, BelRog96]		✓	✓
Chain-based [GCS+17,Sha18]	✓	✓	
HaSS [This work]	✓	✓	✓

For signing and verification: #ROs = 6.

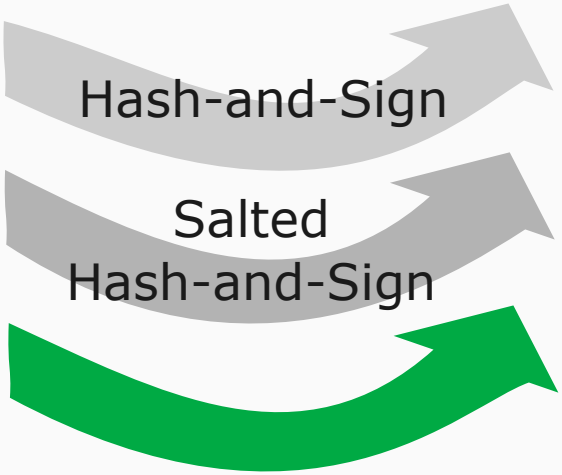
Our Contribution: Hash-and-Subset-Sign (HaSS)



Our Contribution: Hash-and-Subset-Sign (HaSS)







:Loses Tightness

: Loses Uniqueness

: Preserves Tightness and Uniqueness

Methods

Methods Outline

We will discuss these in order:

- Hash-and-Sign Transformation (Unique).
- Salted Hash-and-Sign Transformation (Tight).
- HaSS (Tight and Unique).

Given

A unique signature scheme with tight RUF-RMA security. Signing algorithm: $\mathsf{Sign}^{\mathsf{in}}(\mathsf{sk}, m) = \sigma$.

Hash-and-Sign (FDH)

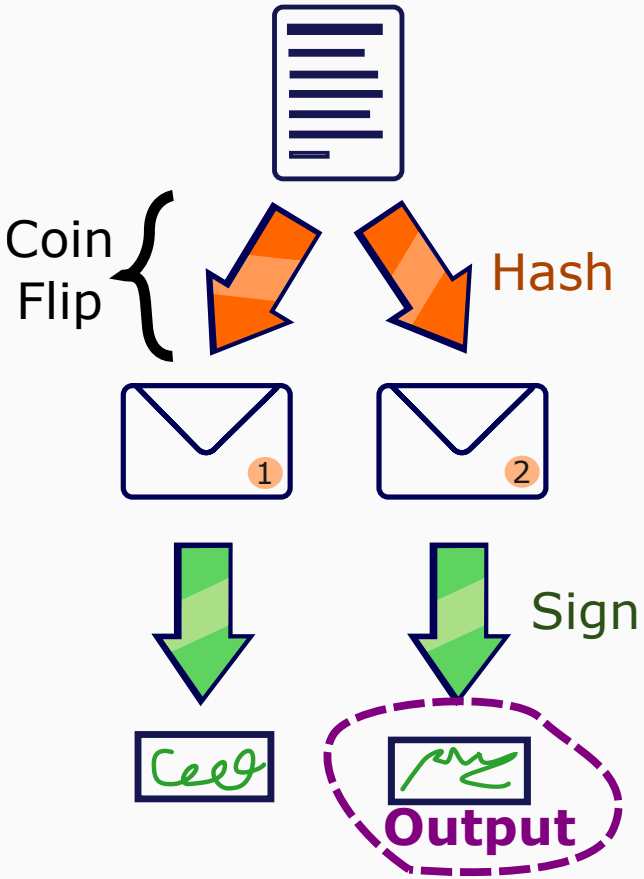


Scheme

$\mathsf{Sign}(\mathsf{sk}, m) = \mathsf{Sign}^{\mathsf{in}}(\mathsf{sk}, \mathcal{H}(m))$.

- Unique.
- Not Tight: $\ell_{\mathcal{R}}(\lambda) = \Omega(q_{\mathsf{Sign}})$.
- Idea: Program random oracle.
- Reduction either knows the signature or can use it to solve underlying problem, but not both.
- Guessing argument.

Katz and Wang (PSS)



Scheme

$\mathsf{Sign}(\mathsf{sk}, m) = \mathsf{Sign}^{\mathsf{in}}(\mathsf{sk}, \mathcal{H}(m||b))$,
$b \leftarrow\!\!\$\ \{0,1\}$.

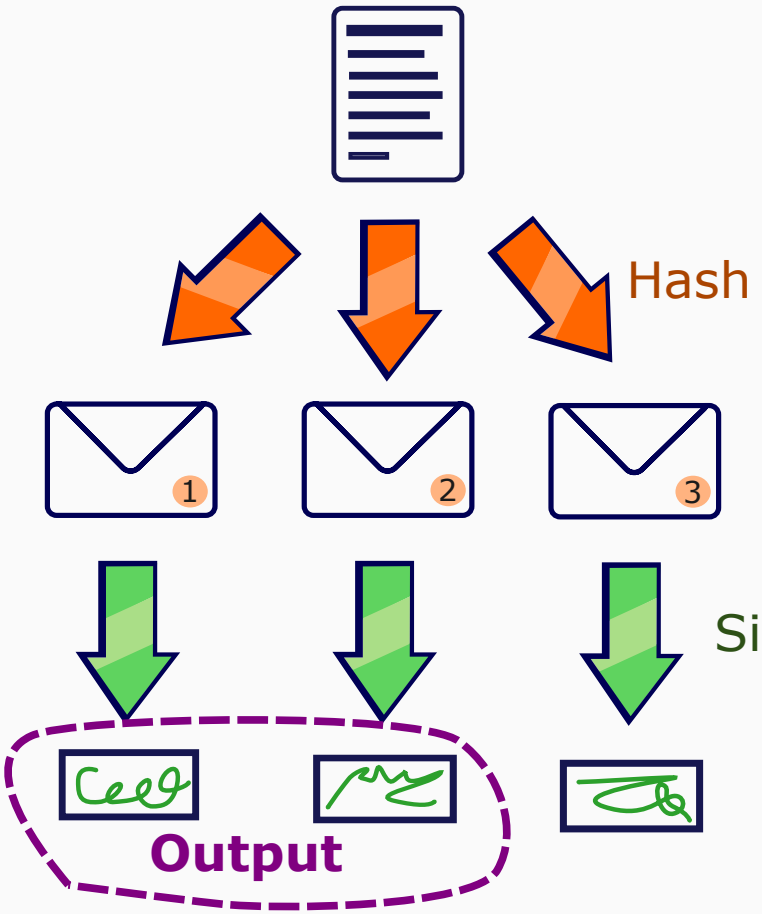
- Not Unique
- Tight: $\ell_{\mathcal{R}}(\lambda) = 2$.
- Reduction knows one of the two signatures
- Adversary gives the other with probability $1/2$.

Idea

Our Scheme: Core Idea

- Trick the adversary into doing work that the reduction cannot do itself.
- $\Longrightarrow$ A Signing algorithm with constant probability to do "too much work".

Our Scheme



Split cases based on $\mathcal{H}'(\sigma) \in \{0,1\}$
$b = (\mathcal{H}'(\sigma_1), \mathcal{H}'(\sigma_2), \mathcal{H}'(\sigma_3))$

a) If $b = 0^3 \vee b = 1^3$ output $(\sigma_1, \sigma_2, \sigma_3)$.

b) If $b = (0, 1, ?)$ output $(\sigma_1, \sigma_2, \perp)$.

c) If $b = (?, 0, 1)$ output $(\perp, \sigma_2, \sigma_3)$.

d) If $b = (1, ?, 0)$ output $(\sigma_1, \perp, \sigma_3)$.

- Uniqueness: Each case is verifiable and mutually exclusive
- Tight: $\ell_{\mathcal{R}}(\lambda) = 4$.

Conclusion

Main Result: Generic transformation from RUF-RMA to EUF-CMA which preserves tightness and uniqueness.

Extensions to Multi-User Security with different drawbacks:

- Either only allows for semi-adaptive corruptions,
- or scheme parameters scale polynomially in q_{Sign}.

Future Work:

- Full Multi-User Security.
- Quantum instantiation + QROM.
- Smaller Signatures.

Thank you for listening!